



A Level • Cambridge (CIE) • Chemistry

30 mins

16 questions

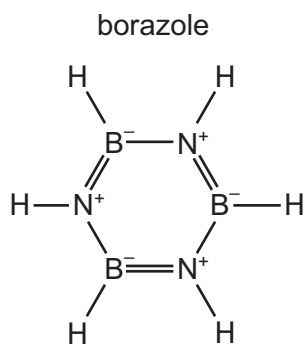
Multiple Choice Practice

Topical Practice MCQ

3. Chemical bonding

Carefully selected past paper questions by topic.

- 6 A structure for borazole, $\text{N}_3\text{B}_3\text{H}_6$, is shown.



Which shape is borazole and how many π electrons are there in the structure?

	shape	number of π electrons
A	non-planar	3
B	non-planar	6
C	planar	3
D	planar	6

14 In a sample of pure water, what is the maximum number of hydrogen bonds that one molecule of water can be involved in?

A 1

B 2

C 3

D 4

22 The boiling points of Br₂, ICl and IBr are given in the table.

	Br ₂	ICl	IBr
boiling point/ °C	59	97	116

Which row explains:

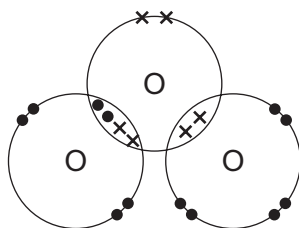
- why the boiling point of ICl is greater than Br₂
- why the boiling point of IBr is greater than ICl?

	boiling point of ICl is greater than Br ₂	boiling point of IBr is greater than ICl
A	ICl has stronger instantaneous dipole-induced dipoles	IBr has stronger instantaneous dipole-induced dipoles
B	ICl has permanent dipoles	IBr has stronger instantaneous dipole-induced dipoles
C	ICl has stronger instantaneous dipole-induced dipoles	IBr has stronger permanent dipoles
D	ICl has permanent dipoles	IBr has stronger permanent dipoles

- 5** In which structure are three atoms bonded together in a straight line?
- A** poly(ethene), $-(\text{CH}_2\text{CH}_2)_n-$
 - B** propane, C_3H_8
 - C** silicon tetrachloride, SiCl_4
 - D** sulfur hexafluoride, SF_6

13 VSEPR theory should be used in answering this question.

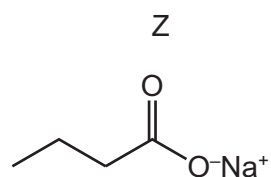
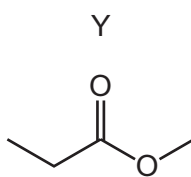
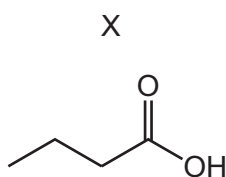
The dot-and-cross diagram for an ozone, O_3 , molecule is shown.



What is the predicted bond angle in this molecule?

- A** 107° **B** 109.5° **C** 117° **D** 120°

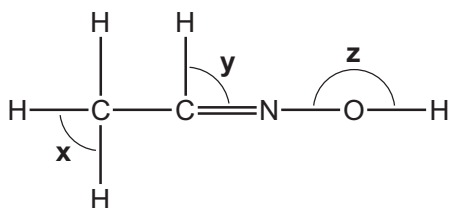
- 3 The structures represent three compounds, each with four carbon atoms per molecule.



Which row is correct?

	lowest boiling point	→	highest boiling point
A	X	Y	Z
B	Y	X	Z
C	Z	X	Y
D	Z	Y	X

- 4 Ethanal reacts with hydroxylamine, NH_2OH , to form the molecule shown.



What is the order of **increasing** bond angle in this structure from smallest to largest?

- A** z, x, y **B** y, z, x **C** x, z, y **D** z, y, x

- 7 The strength of hydrogen bonding increases as the electronegativity of the element bonded to hydrogen increases.

Some information for a range of hydrides is given.

hydride	boiling point /K
PH ₃	185
HCl	188
HF	293
H ₂ O	373

Which statement and reason about these hydrides is correct?

- A** The boiling point of PH₃ is much lower than the boiling point of H₂O because PH₃ does not form hydrogen bonds or instantaneous dipole-induced dipole forces between its molecules.
- B** The boiling point of HF is higher than the boiling point of HCl because the bond energy of H–F is greater than the bond energy of H–Cl.
- C** The boiling point of H₂O is higher than the boiling point of HF because each hydrogen bond between the H₂O molecules is stronger than each hydrogen bond between HF molecules.
- D** The boiling points of PH₃ and HCl are similar because the molecules of PH₃ and HCl have the same number of electrons and similar intermolecular forces.

4 In which set do all the molecules have all their atoms arranged in one plane?

A $AlCl_3$, BF_3 , PH_3

B $AlCl_3$, CO_2 , NH_3

C BF_3 , C_2H_4 , C_3H_6

D C_2H_4 , CO_2 , H_2O

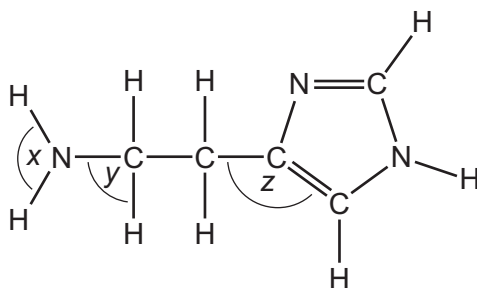
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- 6 Histamine is produced in the body to help fight infection. Its shape allows it to fit into receptors which expand blood vessels.

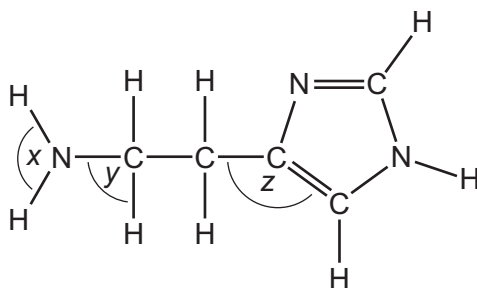


histamine

What are the bond angles x , y and z in histamine, from the smallest to the largest?

	smallest bond angle	→	largest bond angle
A	x	y	z
B	y	x	z
C	y	z	x
D	z	y	x

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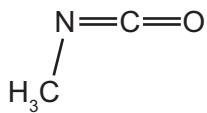
	smallest bond angle	→	largest bond angle
A	x	y	z
B	y	x	z
C	y	z	x
D	z	y	x

- 5 Methyl isocyanate, CH_3NCO , is a toxic liquid which is used in the manufacture of some pesticides.

In the methyl isocyanate molecule, the sequence of atoms is $\text{H}_3\text{C}-\text{N}=\text{C}=\text{O}$.

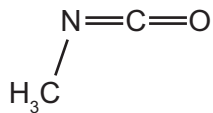
What is the approximate angle between the bonds formed by the N atom?

A



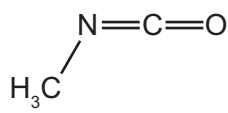
104°

B



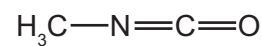
109°

C



120°

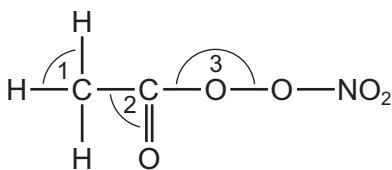
D



180°

- 7 Organic nitrates in photochemical smog can cause breathing difficulties.

The diagram shows an example of an organic nitrate molecule.

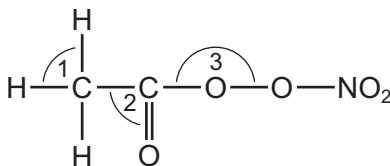


What is the correct order of the bond angles shown in ascending order (smallest first)?

- A** $1 \rightarrow 2 \rightarrow 3$ **B** $2 \rightarrow 1 \rightarrow 3$ **C** $3 \rightarrow 1 \rightarrow 2$ **D** $3 \rightarrow 2 \rightarrow 1$

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Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_13	3. Chemical bonding	D
2	9701_m23_qp_12	3. Chemical bonding	D
3	9701_s23_qp_11	3. Chemical bonding	B
4	9701_s22_qp_11	3. Chemical bonding	D
5	9701_w22_qp_12	3. Chemical bonding	C
6	9701_s21_qp_12	3. Chemical bonding	B
7	9701_s21_qp_13	3. Chemical bonding	A
8	9701_w21_qp_12	3. Chemical bonding	D
9	9701_w18_qp_12	3. Chemical bonding	D
10	9701_w17_qp_11	3. Chemical bonding	D
11	9701_w17_qp_13	3. Chemical bonding	D
12	9701_w16_qp_11	3. Chemical bonding	A
13	9701_w16_qp_13	3. Chemical bonding	A
14	9701_w11_qp_13	3. Chemical bonding	C
15	9701_w10_qp_11	3. Chemical bonding	C
16	9701_w10_qp_13	3. Chemical bonding	C